

Introduction to Computational Thinking and Programming with Python

Instructor: Sarom Leang, PhD (Teacher/Professor)

Specialist: Crystal Rafael-Castelan

Innovation Room 203



Pronouns

- Instructor: Sarom Leang, Ph.D. (Instructor/Teacher)
- Specialist: Crystal Rafael-Castelan



Schedule

- 10:00 AM – 10:15 AM (Homeroom)
- 10:15 AM – 11:35 AM (G1)
- 11:40 AM – 12:35 PM (Lunch)
- 12:40 PM – 02:00 PM (G2)



Student Expectations

- **NO FOOD**
- **NO DRINKS** (on the table)
- Be respectful to individuals and property
- Be open to learning
- Be open to not understanding
- Be patient with yourself
- Ask questions
- Explore
- **Embrace failure**



Course Overview

- This course introduces the **fundamental** building blocks of **computational thinking** and **computer programming** using the **Python** language.
- Upon successful completion of this course, students will be able to:
 - Improve their problem-solving skills
 - Write, read, and execute Python code using basic data types and operators



Course Overview (cont.)

Exploratory Topics

- Session #1 – November 1, 2025
 - Entrance Survey
 - How Computers Work
- Session #2 – December 6, 2025
 - Parallel Computing
- Session #3 – February 7, 2026
 - Generative Artificial Intelligence
- Session #4 – March 7, 2026
 - Quantum Computing
- Session #5 – April 11, 2026
 - Future of Computing
 - Exit survey

Topics

- Algorithms and Problem Solving
- Debugging and Troubleshooting
- Loops
- Algorithms and Syntax
- Variables and Conditionals
- Variable Arithmetic
- Conditionals (If/Else)
- Compound Conditionals



Sarom Leang, PhD (Instructor/Teacher)

- 1982 Coming to America!
Escaped the Khmer Rouge Genocide
- 1998 EIP Class 7 Scholar
Wakefield High School
- 2004 B.S. Chemistry (Honors, High Distinction), Minor CS
George Mason University
- 2011 Ph.D. Physical Chemistry / Postdoctoral Researcher
Iowa State University
- 2014 Assistant Research Scientist
The Ames Laboratory, Department of Energy
- 2018 Senior Computational Scientist/Senior Software Engineer
EP Analytics, Inc.
- 2025 High Performance Computing Team Lead
???



Entrance Survey (G1)

- Visit the following URL and answer the survey as best you can:
 - G1 – <https://bit.ly/2025G1>



Entrance Survey (G2)

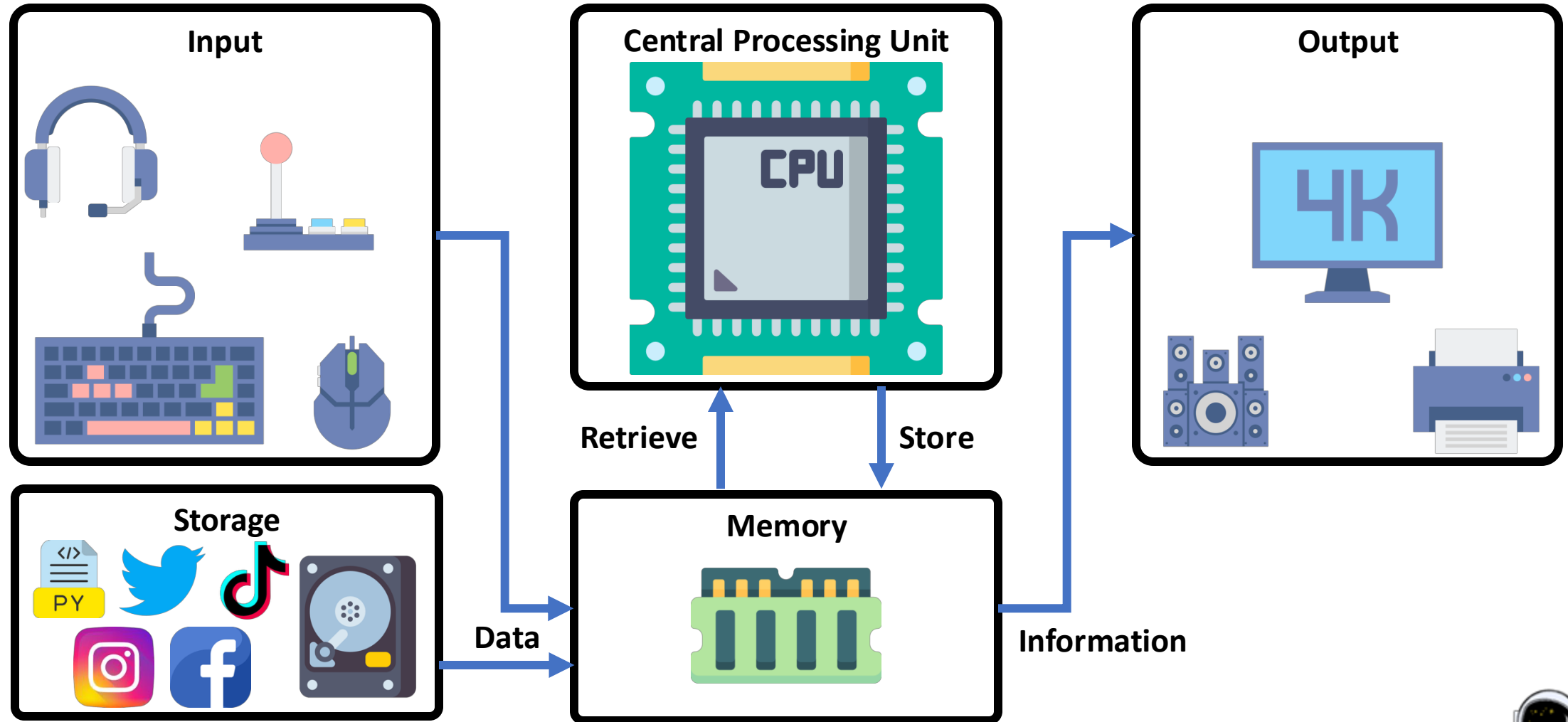
- Visit the following URL and answer the survey as best you can:
 - G2 – <https://bit.ly/2025G2Entrance>



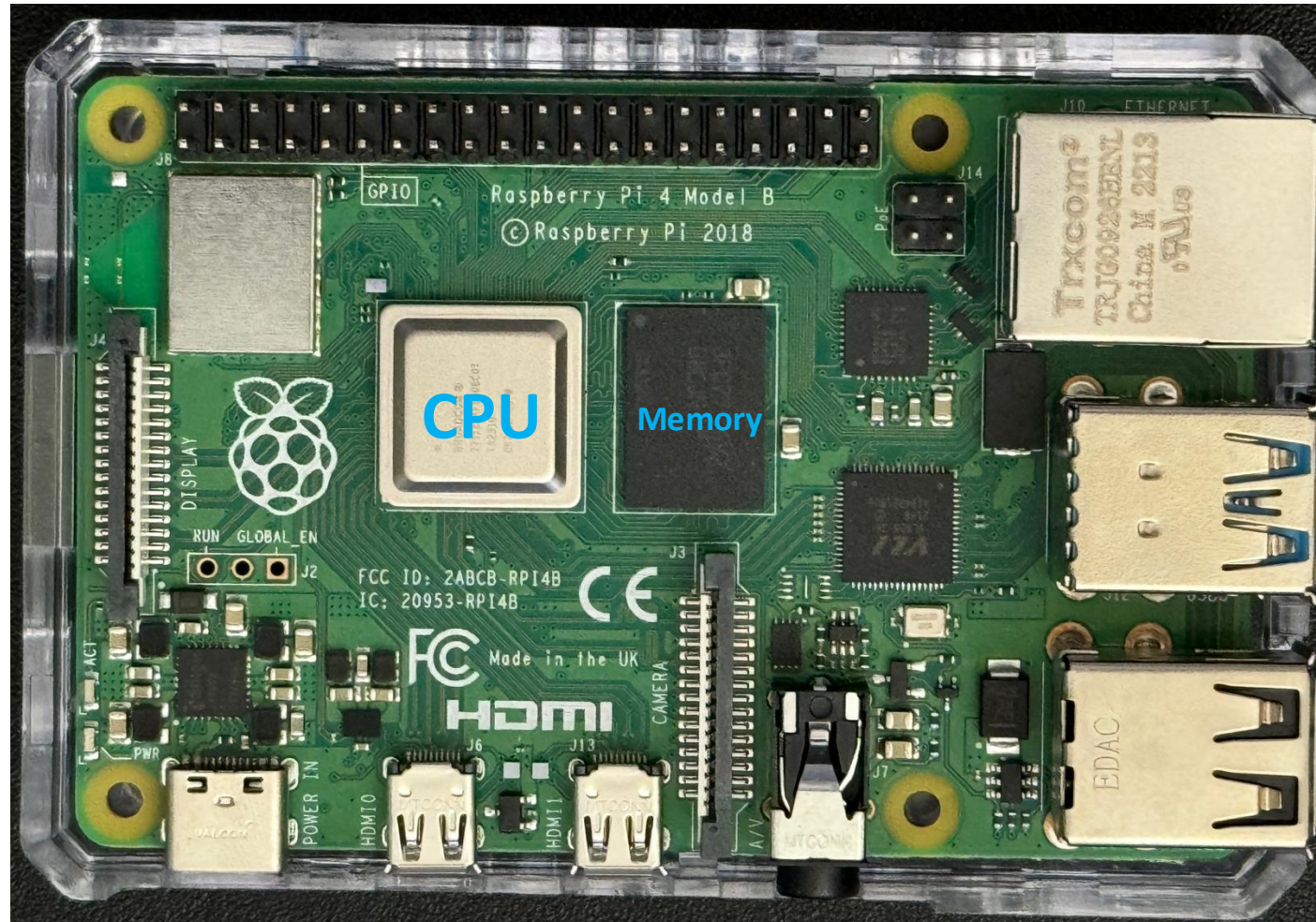
Discussion – How Do Computers Work?



Discussion – How Do Computers Work?

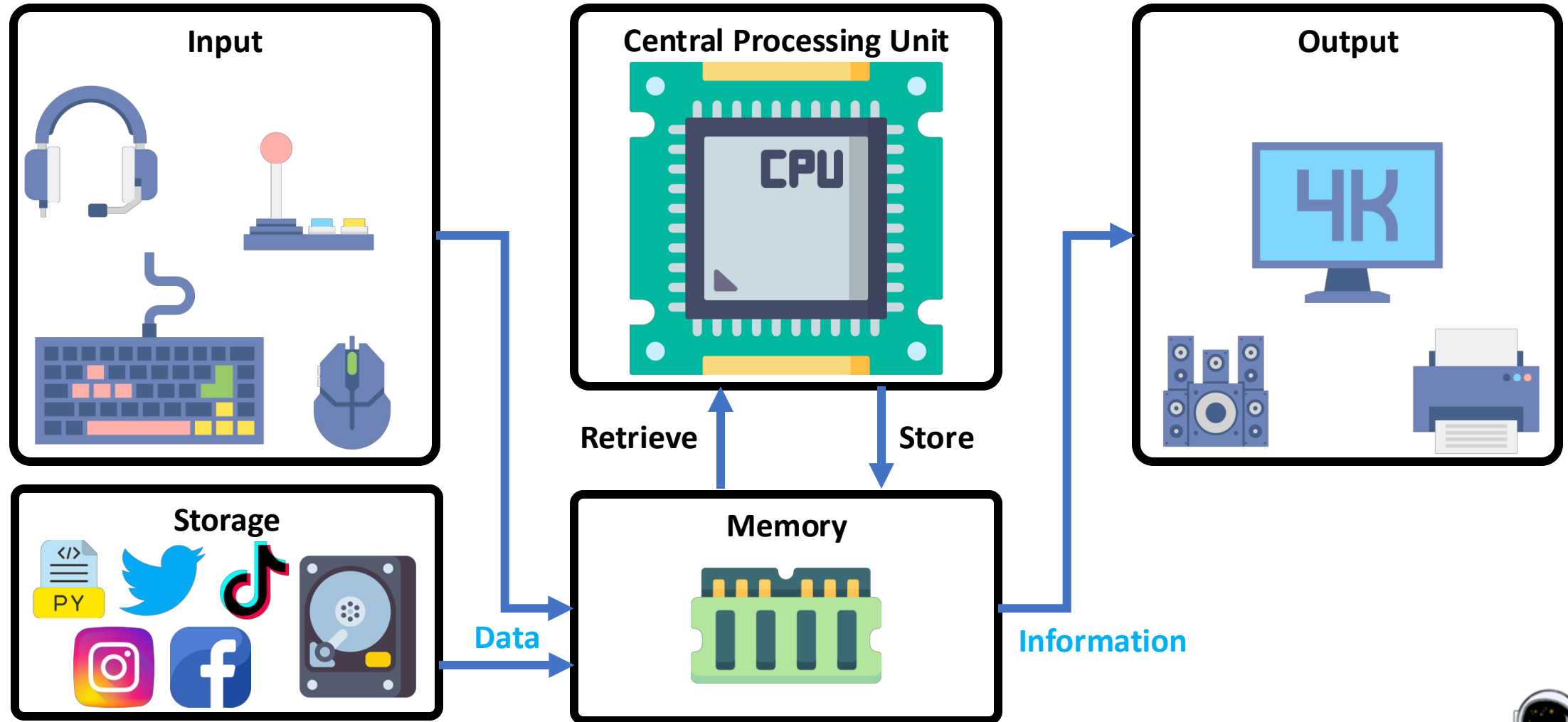


Discussion – How Do Computers Work?



Raspberry Pi

Discussion – How Do Computers Work?



Data vs. Information

- 79, 68, 68, 70, 61, 64, 68, 81, 82, 79, 64



Data vs. Information

October 23	79
October 24	68
October 25	68
October 26	70
October 27	61
October 28	64
October 29	68
October 30	81
October 31	82
November 1	79
November 2	64
Average	71.3



Data vs. Information

2024	High Temps (°F)
October 23	79
October 24	68
October 25	68
October 26	70
October 27	61
October 28	64
October 29	68
October 30	81
October 31	82
November 1	79
November 2	64
Average	71.3



Data vs. Information

2024	Site Visitors (#)
October 23	79
October 24	68
October 25	68
October 26	70
October 27	61
October 28	64
October 29	68
October 30	81
October 31	82
November 1	79
November 2	64
Average	71.3



Data vs. Information

2024	Quiz Scores (%)
October 23	79
October 24	68
October 25	68
October 26	70
October 27	61
October 28	64
October 29	68
October 30	81
October 31	82
November 1	79
November 2	64
Average	71.3



Data vs. Information

2024	Stock Price (\$)
October 23	79
October 24	68
October 25	68
October 26	70
October 27	61
October 28	64
October 29	68
October 30	81
October 31	82
November 1	79
November 2	64
Average	71.3



Discussion – How Do Computers Work?

Data

- collection of raw statistics or facts
- unorganized
- does not inherently have meaning
- does not offer a clear enough picture to make decisions

Information

- offers context to those facts, statistics
- is organized
- has purpose due to analysis
- **with context, information can help influence decisions**



Discussion – How Do Computers Work?

Data

- High temperatures
- Site visitors
- Student scores
- Stock prices

Information

- What clothes to pack for a trip
- Best day to drop new products
- Adjust lesson plans
- Determine impact earnings report



Discussion – What is a computer?

- An electronic device that **stores, retrieves, and processes** data.



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- A **programmable** electronic device that **stores, retrieves, and processes** data.



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- A **programmable** electronic device that **stores, retrieves, and processes** data following a **set of instructions**.



Discussion – What is a computer?

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program



Discussion – What is computer programming?



Discussion – What is computer programming?

- The **mental process** of developing a **set of instructions** for a computer.
 - Design and planning



Discussion – What is computer programming?

- The **mental process** of developing a **set of instructions** for a computer.
 - Design and planning
- What is the **physical process** called?



Discussion – What is computer programming?

- The **mental process** of developing a **set of instructions** for a computer.
 - Design and planning
- What is the **physical process** called?
 - Coding



Example – Set of Instructions (For Humans)

- Take one step to the left
- Step backward
- Jump once in place with both feet
- Stomp your right foot in place
- Stomp your left foot in place
- Rock step backward on right foot, then do three quick steps in place
- Jump and cross your legs by moving the right leg in front of the left
- Jump again to uncross your legs



Example – Set of Instructions (For Computers)

```
r=int(input("Enter upper limit: "))
for a in range(2,r+1):
    k=0
    for i in range(2,a//2+1):
        if(a%i==0):
            k=k+1
    if(k<=0):
        print(a)
```



Discussion – Why learn computer programming?



Discussion – Why learn computer programming?

- Improve your problem-solving skills
- Process large amounts of data (finance, manufacturing, healthcare, science)
- Create/design a web site/application/game
- To better understand computer programs/technology
- As a career (Glassdoor salary search for Washington D.C. area)
 - Software Developer - \$109,099/year
 - Data Scientist - \$118K/year
 - Cyber Security Engineer - \$136K/year
 - Machine Learning Engineer - \$141K/year
 - Software Architect - \$158K/year
 - Software Manager – \$163K/year
- As a hobby



Discussion - What is the impact of computer programming (e.g., world, daily life)?



Discussion - What is the impact of computer programming (e.g., world, daily life)?

- Artificial intelligence/machine learning
- Robotics/automation
- Cybersecurity/threats
- Websites/E-commerce
- Payment processing/transactions
- Mobile Apps
- Weather modeling
- Global positioning system (GPS)
- Data encryption
- Schedule/inventory management
- Social media





OZARIA

Lesson Plan

Chapter 1: Sky Mountain



Logging into Ozaria (G1)

1. Go to: <https://www.ozaria.com>
2. Click “**Sign Up**”
3. Enter in your **class code** →
4. **Create an account** or login with your Google account
5. Make sure to **write down your user name!**
6. Click **Continue** to start playing →

Students, request a Class Code from your Teacher to create an Account!

Enter Class Code:

PushZooEat

Create Student Account



Intro to CS (Python)

Teacher: Charlotte Cheng

Prologue: Sky Mountain [view map](#)

[view my classmates' projects](#)

Continue

0.0%

Logging into Ozaria (G1) - Javascript

1. Go to: <https://www.ozaria.com>
2. Click “**Sign Up**”
3. Enter in your **class code** →
4. **Create an account** or login with your Google account
5. Make sure to **write down your user name!**
6. Click **Continue** to start playing →

Students, request a Class Code from your Teacher to create an Account!

Enter Class Code:

FaceFunnyCloud

Create Student Account



Intro to CS (Python)

Teacher: Charlotte Cheng

Prologue: Sky Mountain [view map](#)

[view my classmates' projects](#)

Continue

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Logging into Ozaria (G2)

1. Go to: <https://www.ozaria.com>
2. Click “**Sign Up**”
3. Enter in your **class code** →
4. **Create an account** or login with your Google account
5. Make sure to **write down your user name!**
6. Click **Continue** to start playing →

Students, request a Class Code from your Teacher to create an Account!

Enter Class Code:

FreeLockBoat

Create Student Account



Intro to CS (Python)

Teacher: Charlotte Cheng

Prologue: Sky Mountain [view map](#)

[view my classmates' projects](#)

Continue

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Logging into Ozaria (G2) - Javascript

1. Go to: <https://www.ozaria.com>
2. Click “**Sign Up**”
3. Enter in your **class code** →
4. **Create an account** or login with your Google account
5. Make sure to **write down your user name!**
6. Click **Continue** to start playing →

Students, request a Class Code from your Teacher to create an Account!

Enter Class Code:

WinPushFalse

Create Student Account



Intro to CS (Python)

Teacher: Charlotte Cheng

Prologue: Sky Mountain [view map](#)

[view my classmates' projects](#)

Continue

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Using Ozaria Lesson Slides

You can use our turnkey lesson slides to **introduce coding concepts** to

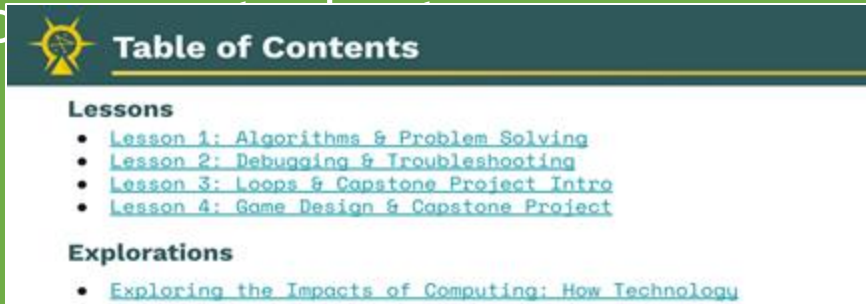


Table of Contents
Lessons
Lesson 1: Algorithms & Problem Solving
Lesson 2: Debugging & Troubleshooting
Lesson 3: Loops & Capstone Project Intro
Lesson 4: Game Design & Capstone Project
Explorations
Exploring the Impacts of Computing: How Technology

Each lesson slide deck starts with a **Table of Contents** which includes links so that you can quickly navigate to specific lessons and resources.



Algorithms

a sequence of instructions that can be used to solve a problem or set of problems

- Highlight the problems shown on the slide: baking cookies, building a house, solving a maze, and writing a program.
 - Then ask students how these problems can be broken into a sequence of steps.
- For example:
 - baking cookies = measure ingredients, mix ingredients, form balls with dough, bake
 - building a house = build the foundation, the walls, the roof, interior
- Highlight how coding is just creating algorithms to solve problems

Be sure to check the **speaker notes** to find specific talking points for each slide along with suggested solutions and answers.

Using Ozaria Lesson Slides

The slides also guide students during **independent practice** and offer **unplugged extension activities**.

Independent Practice

 **Log into Ozaria**
<https://www.ozaria.com>

 **Play Ozaria**
Start at **Intro: Up the Mountain** and stop after you finish **Cutscene: The Spirit Lands**

 **Extension Activity: Design a Map**
Use the link provided by your teacher to complete this activity.

 **Need Help?**
Always try it once & try using the troubleshooting guide first. If you need help after that, then reach out to a classmate or the teacher.

p. of independent practice

Pacing
Click Progress page, go to Chapter 1 and lock the Intro: Fixing the Light
should first play Ozaria through the Spirit Lands Cutscene (video). If they finish early, they can work on the Extension Activity
information on locking and pacing, see: https://docs.google.com/presentation/d/1JAs_w8RjqtCtCQRGGI2_pRzZGar8rcNjEBcSFUYMF6c/edit?usp=sharing

Each lesson includes an **Independent Practice** slide that recommends where students should start and stop playing in Ozaria.

This slide also includes an **Extension Activity** that students can try if they finish early.

Check the **Speaker Notes** for links to Extension Activity content and instructions on how to lock or unlock levels for your students.

Make a Copy of these Lesson Slides

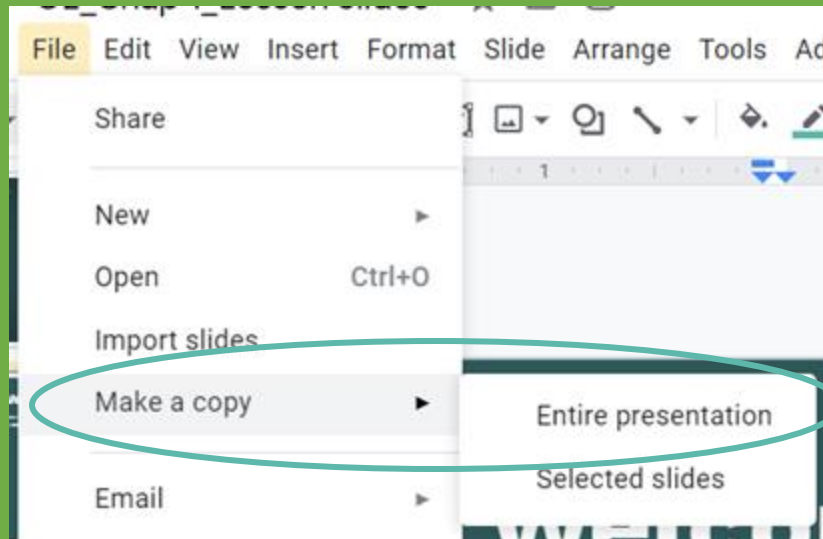
If you would like to **edit and adapt** these slides to the needs of your school or class, you can **make a copy** of them.

1



Click on the **File** menu.

2



Click **Make a copy** and copy the **entire presentation**

Customizing the Lesson Slides

3

Copy document



Name

Copy of Oz_Chap 1_Lesson Slides

4

Folder

Chapter 1

Copy will be created in a shared folder

5

☐ Share it with the same people

☐ Copy comments

☐ Include resolved comments

☐ Remove all speaker notes

Cancel

OK

You can **name** your version of the slides.

Then select the **folder** where you'd like to save your slides.

If you're **sharing the slides with students**, you may want to **remove speaker notes** as they contain solutions to activities

How to pace a lesson. . .



- ❑ Use the **first section** of the lesson slides (up to **Independent Practice slide**) to introduce the new concept for that lesson
- ❑ If applicable, watch any **Cutscene** videos as a whole class to introduce the story line
- ❑ Have students independently play through the suggested levels on the **Independent Practice slide**
 - ❑ This section which will also include Cinematics and Concept Checks
- ❑ **Extension Activities** are found in the **speaker notes** of the **Independent Practice slide**. You can use them to differentiate pacing for fast finishers or for students who need reinforcement.
- ❑ Use the **Check-In** section as an opportunity for students to reflect on what they have learned in the lesson



Lesson 1

Algorithms & Problem Solving



Welcome to the world of

OZARIA



Your journey begins...





Today's Journey



Warm-Up: Algorithms & Sequences



Learn **how to use Ozaria**

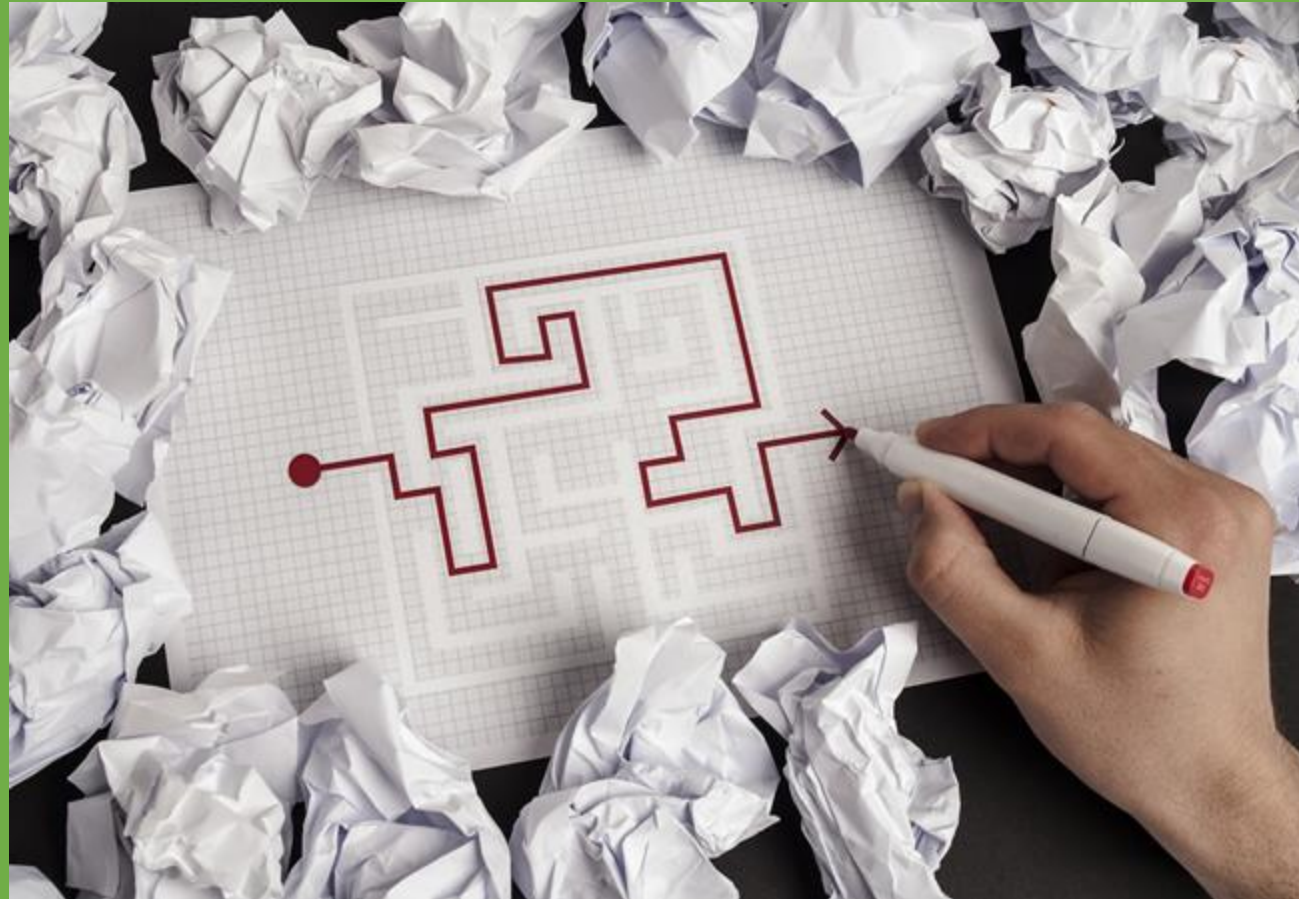


Play Ozaria: Get to the Spirit Lands



Optional Extension: Design a Map

Algorithms & Sequences



Algorithms

a sequence of instructions that can be used to solve a problem or set of problems



```
string sInput;  
int iLength, iN;  
double dblTemp;  
bool again = true;  
  
while (again) {  
    iN = -1;  
    again = false;  
    getline(cin, sInput);  
    system("cls");  
    stringstream(sInput) >> dblTemp;  
    iLength = sInput.length();  
    if (iLength < 4) {  
        again = true;  
        continue;  
    } else if (sInput[iLength - 3] != '.') {  
        again = true;  
        continue;  
    } while (++iN < iLength) {  
        if (isdigit(sInput[iN])) {  
            continue;  
        } else if (iN == (iLength - 3)) {  
            continue;  
        }  
    }  
}
```


Problem Solving with Decomposition

You can solve a complex problem by breaking it down into a sequence of subgoals & steps



What are the subgoals & steps for baking a cake?

Subgoal 1: Make the cake layers

- Measure & combine ingredients
- Make the batter
- Put batter into pans
- Bake the cake layers

Subgoal 2: Make the filling/frosting

- What are the steps?

Subgoal 3: Build the cake

- What are the steps?



Concept Check: Subgoals & Sequences

Write down a sequence of subgoals & steps to help the hero get to the exit stairs.



Subgoal 1: ??

- Move right
- Move right
- ???

Subgoal 2: ??

- Move up
- Move right

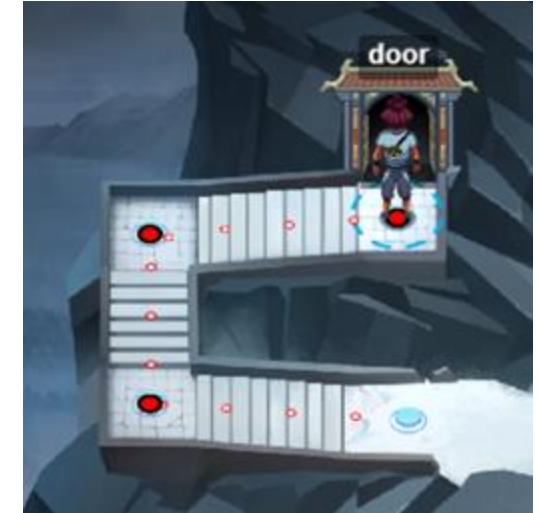
Input & Output

Code is a type of language we use to communicate with computers.

Input



We **input** the code . . .



Output

...and **outputs** actions and/or results based on that code.

the computer processes or reads the code...

Entering Ozaria



Navigating Ozaria



Change **account settings**, **customize** your hero, or go back to your **dashboard**

Use the **map to navigate** to different parts of the game. **Mouse over a stop** to see the **name** of the level.

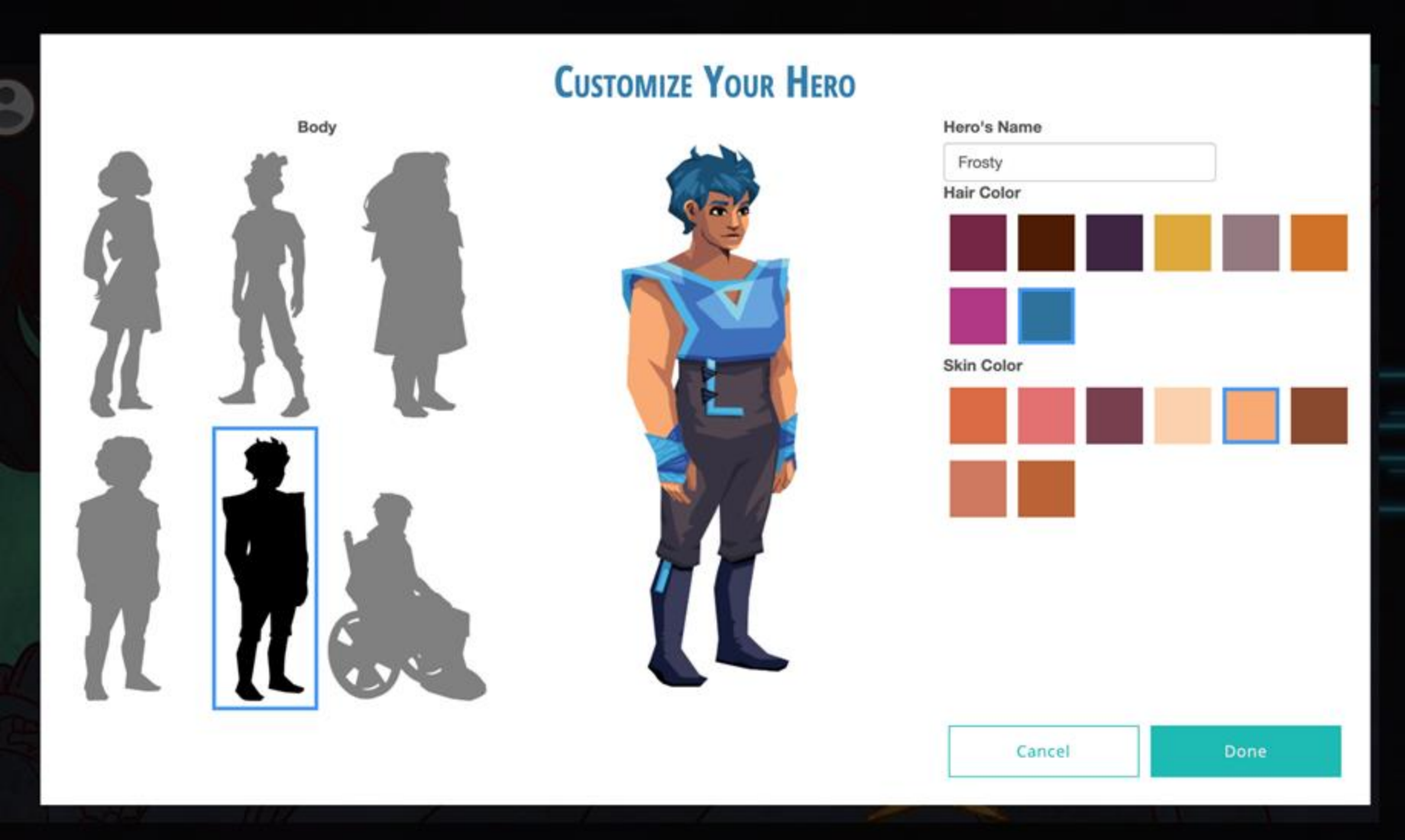
To save space, cinematics and concept checks are organized into **intros**.

Returns to **Map**

Maximizes your Browser

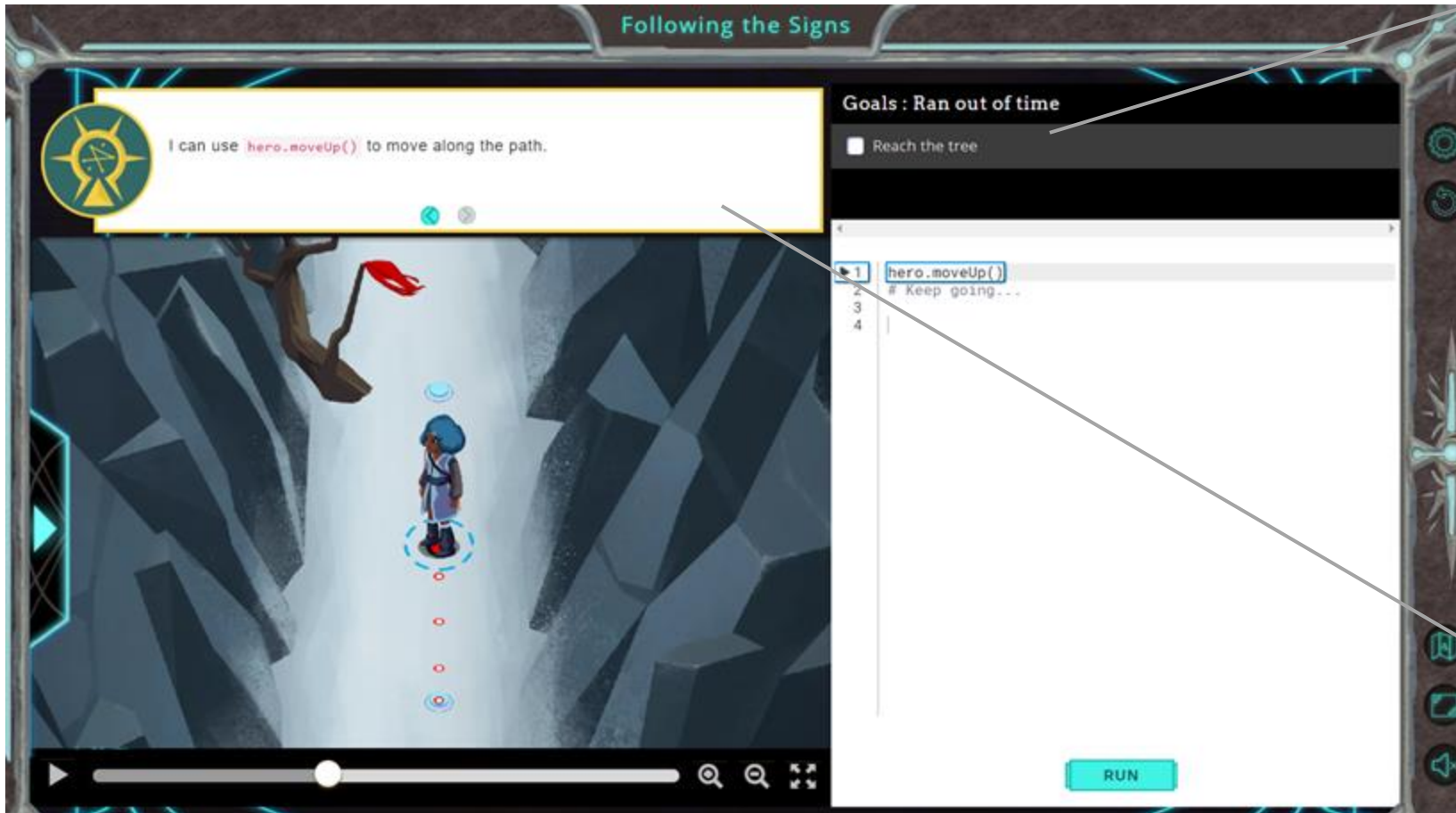
Turn the **Volume** On/Off

Customize Your Hero



Navigating Ozaria

Practice Level: Following the Signs



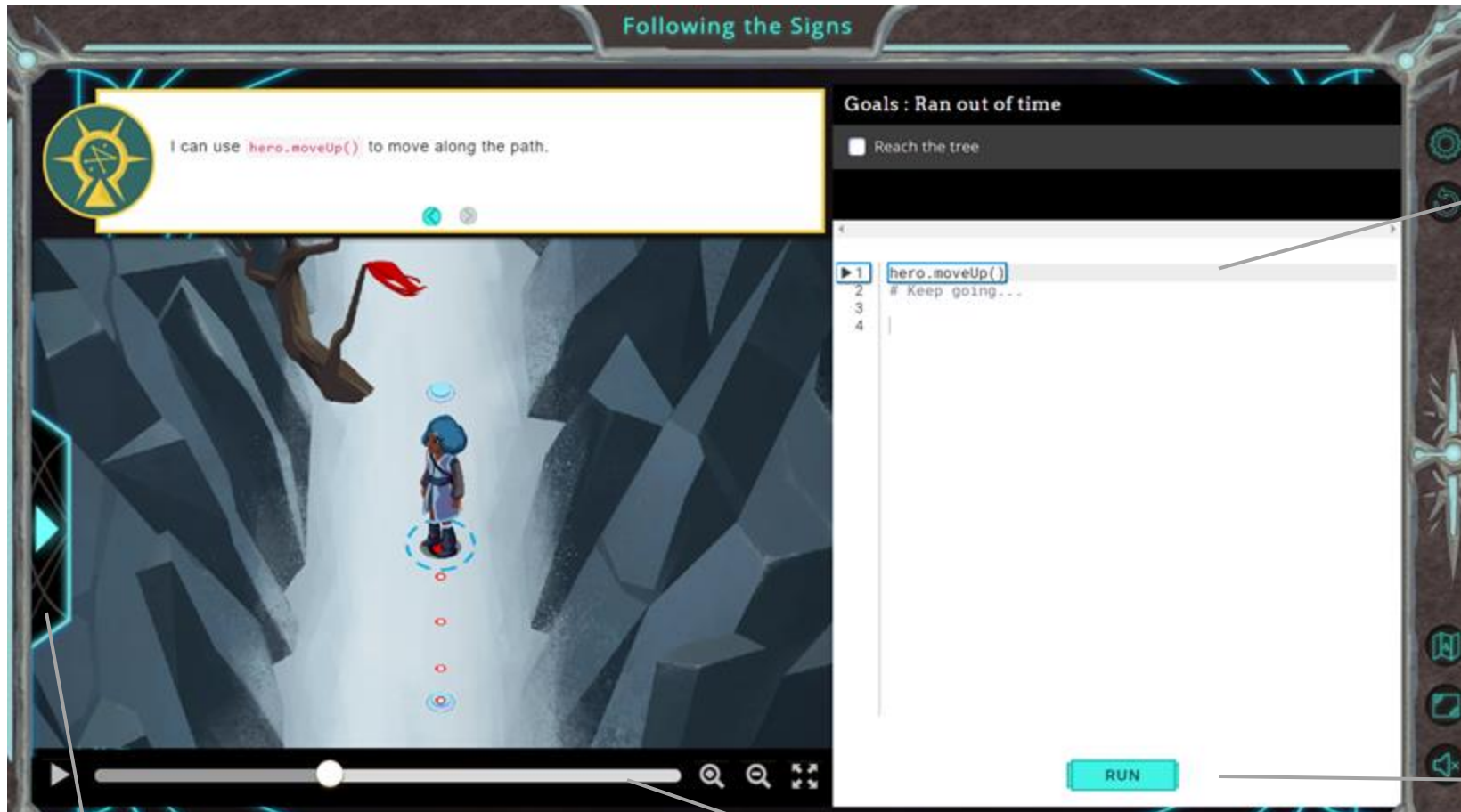
Goals keep track of what you need to do to **complete the level**

Customize your **coding experience**

Restart the level

Use arrows to navigate the tutorial messages. They provide important **hints & instruction.**

Navigating Ozaria



Write your code here

Run your program to see what happens

The **Code Bank** gives you more info about the **different commands** you can use in your code.

Go **backwards and forwards** in your program. You can also **zoom in and out**.

Independent Practice



Check-In





What did you discover?



Chapter 1, Module 1, Lesson 1

What I did:

-How far did you get in Ozaria? What was your favorite part?

What I noticed:

- What did you notice as you played through the game?
- Did you discover or learn something new?

What was challenging?

- Was something confusing about the story or the levels?
- Do you need more practice on a specific part of the code or the game?



Lesson 2

Debugging & Troubleshooting



What did we learn last time?



- What is an **algorithm**?
- What are **sequences**?
- How do you use **decomposition** while **solving problems**?
- What did we do in **Ozaria**?



Today's Journey



Warm-Up: Syntax & Debugging



Customize your **Troubleshooting Guide**



Play Ozaria: Debug helpers in the Spirit Lands



Optional Extension: Design a Debug Level

Syntax & Debugging

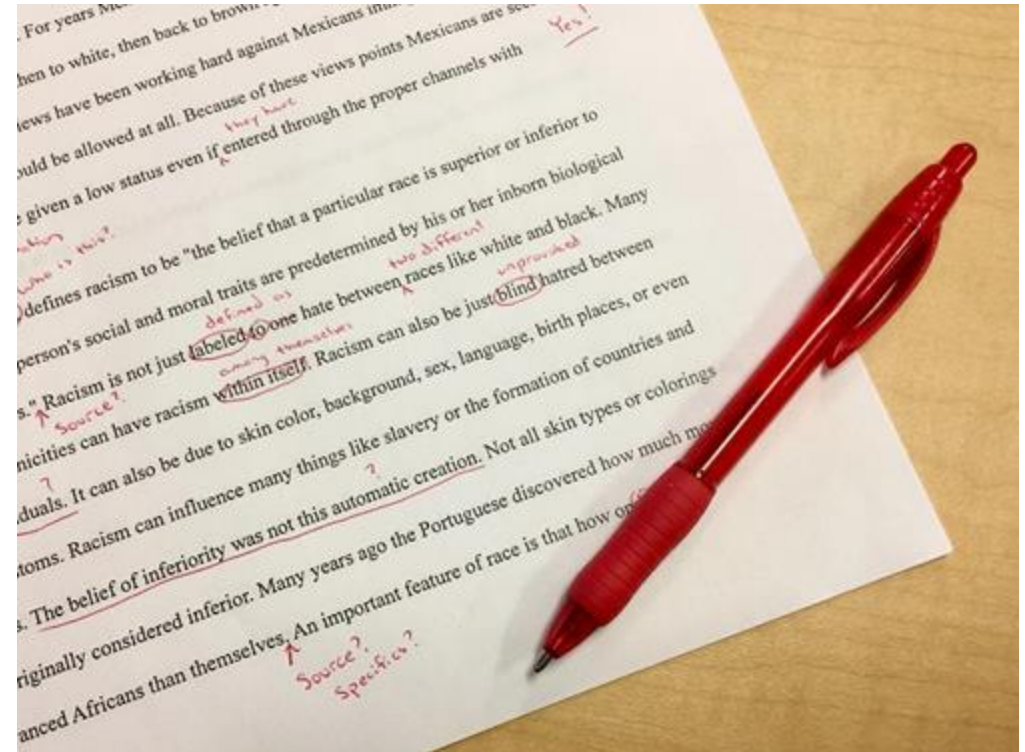


```
✓ 1 helper.moveUp()  
2 helper.build("upArrow")  
▲ 3 help.moveUp()  
4 helper.build("right")  
5 helper.moveRight()  
6 helper.build("upArrow")  
7 helper.upMove()  
8 helper.build("rightArrow")  
9
```

Syntax

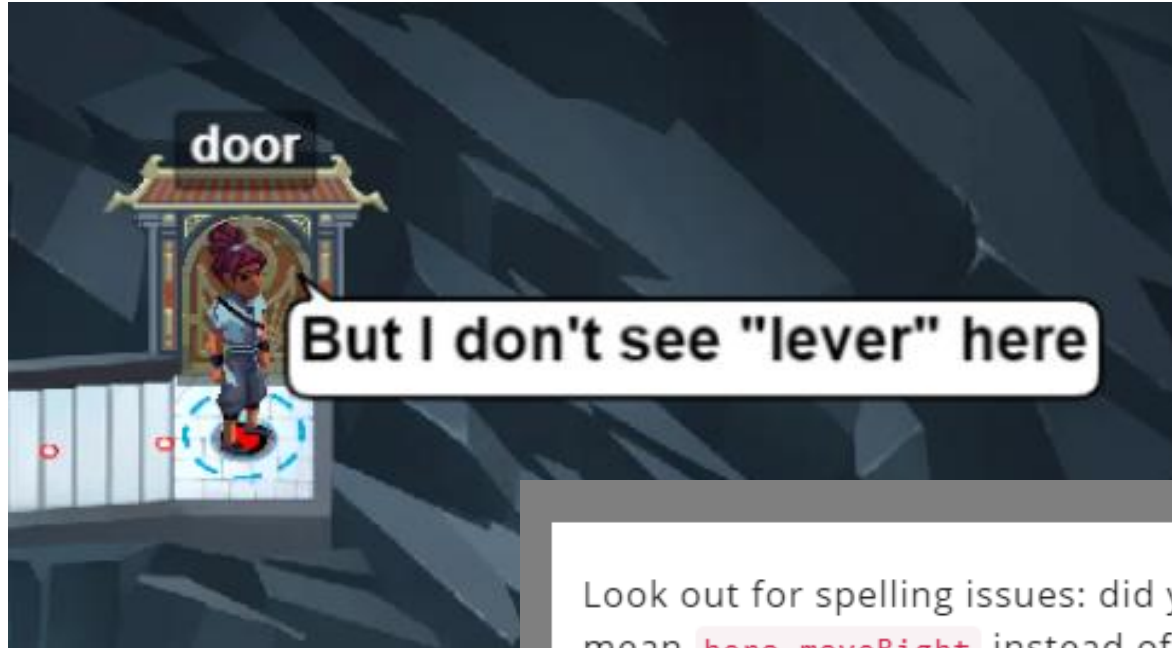
a set of rules for how things are spelled and formatted in code so that the computer can understand it

```
1 hero.moveUp()  
2 hero.move()  
3 hero.moveRight()  
4 hero.moveUp()  
5  
6
```



Debugging

the process of identifying and removing errors from your code so you can meet your goals



Goals : Ran out of time

- ☒ Get to the door
- ☐ Use the door

Look out for spelling issues: did you mean `hero.moveRight` instead of `hero.moveRigt`?

```
✓ 1 hero.moveLeft()  
✓ 2 hero.moveUp()  
⚠ 3 hero.moveRigt()  
4 hero.use("door")  
5
```

Embrace Mistakes

Bugs are a natural and healthy part of coding that lets us learn from our mistakes!



Objects, Methods, and Arguments

In your code, you use objects, methods, and arguments. The upcoming levels will use a different kind of object and different methods



`hero.moveRight(3)`

object

an **object in the game**
that can be controlled or
moved

method

actions that an object
can make in the game

argument

a way you can **customize**
a method

Objects, Methods, and Arguments

The upcoming levels will use a different kind of object and different methods



`helper.build("rightArrow")`

object method argument

The diagram illustrates the components of the code snippet `helper.build("rightArrow")`. Three arrows point from the code to labels below: a green arrow from `helper` to the label `object`, an orange arrow from `build` to the label `method`, and a blue arrow from `"rightArrow"` to the label `argument`.

Check-In





What did you discover?

Chapter 1, Module 1, Lesson 2



What I did:

-How far did you get in Ozaria? What was your favorite part?

What I noticed:

- What did you notice as you played through the game?
- Did you discover or learn something new?



What was challenging?

- Was something confusing about the story or the levels?
- Do you need more practice on a specific part of the code or the game?



Lesson 3

Loops & Capstone Project Intro



What did we learn last time?



- What are **bugs**?
- What's the difference between **syntax bugs** and **logic bugs**?
- What did you add to your **customized troubleshooting guide**?
- What's going on in the story of **Ozaria**?



Today's Journey



Warm-Up: Loops



Learn about your **Capstone Project**



Play Ozaria: Begin your Capstone Project

Loops



Loops

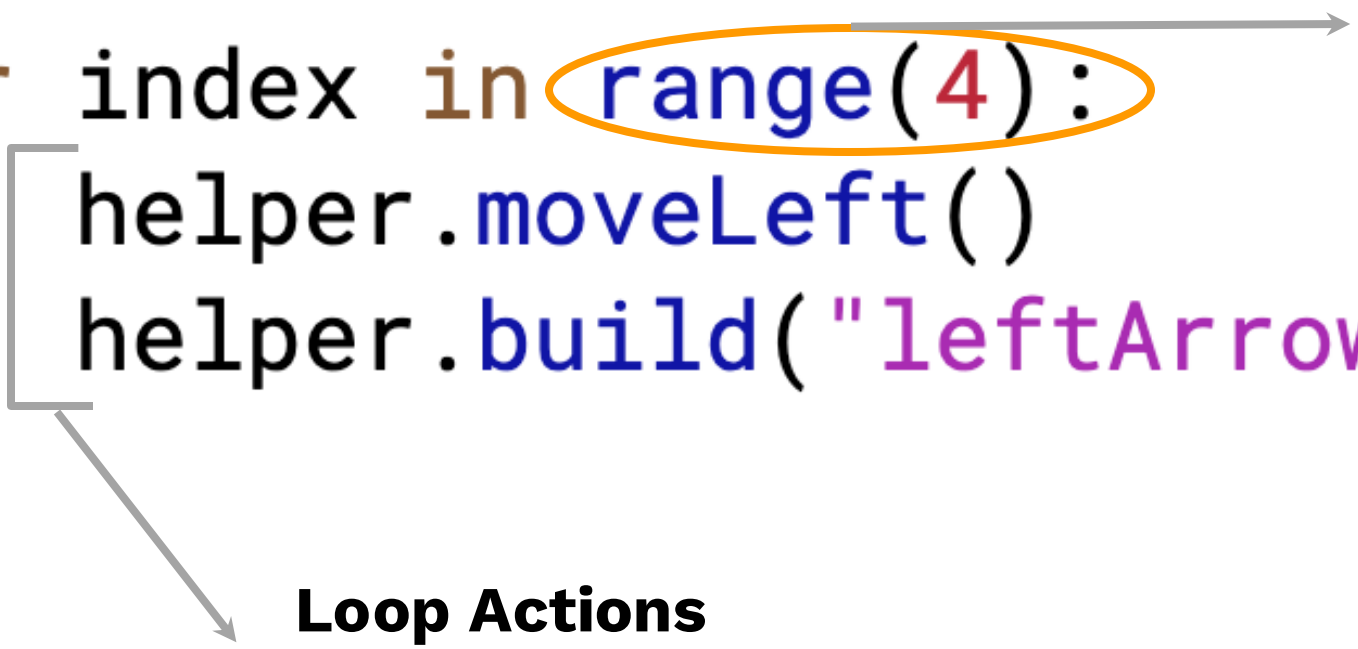
a command that lets us repeat an action multiple times in your code



Coding with Loops (Python)

Loops can make your code more efficient when you have steps that need to repeat over and over again.

```
for index in range(4):  
    helper.moveLeft()  
    helper.build("leftArrow")
```



Repeat Number

sets the number of times the loop repeats

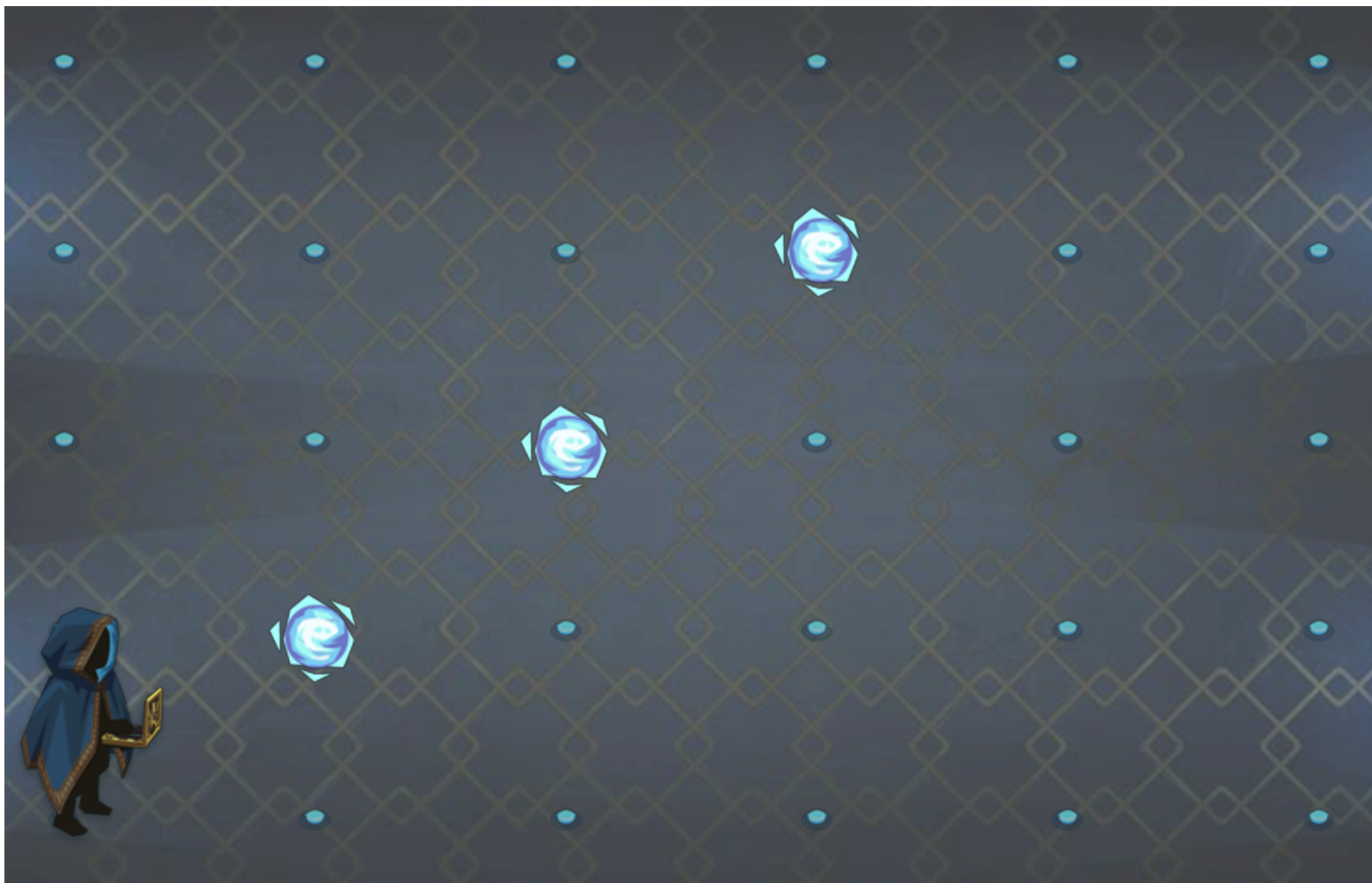
Loop Actions

All the **indented** actions inside the loop will repeat each time the loop repeats



Concept Check: Loops

Write down a repeating sequence of commands that lets the hero collect all the totems



Repeat X Times

-hero moves . . .

Capstone Project

Goal: Design a game to fight the Darkness!



Subgoals

- Use sequences & loops
- Add obstacles and other items to your game
- Customize your game
- Get feedback and revise your game

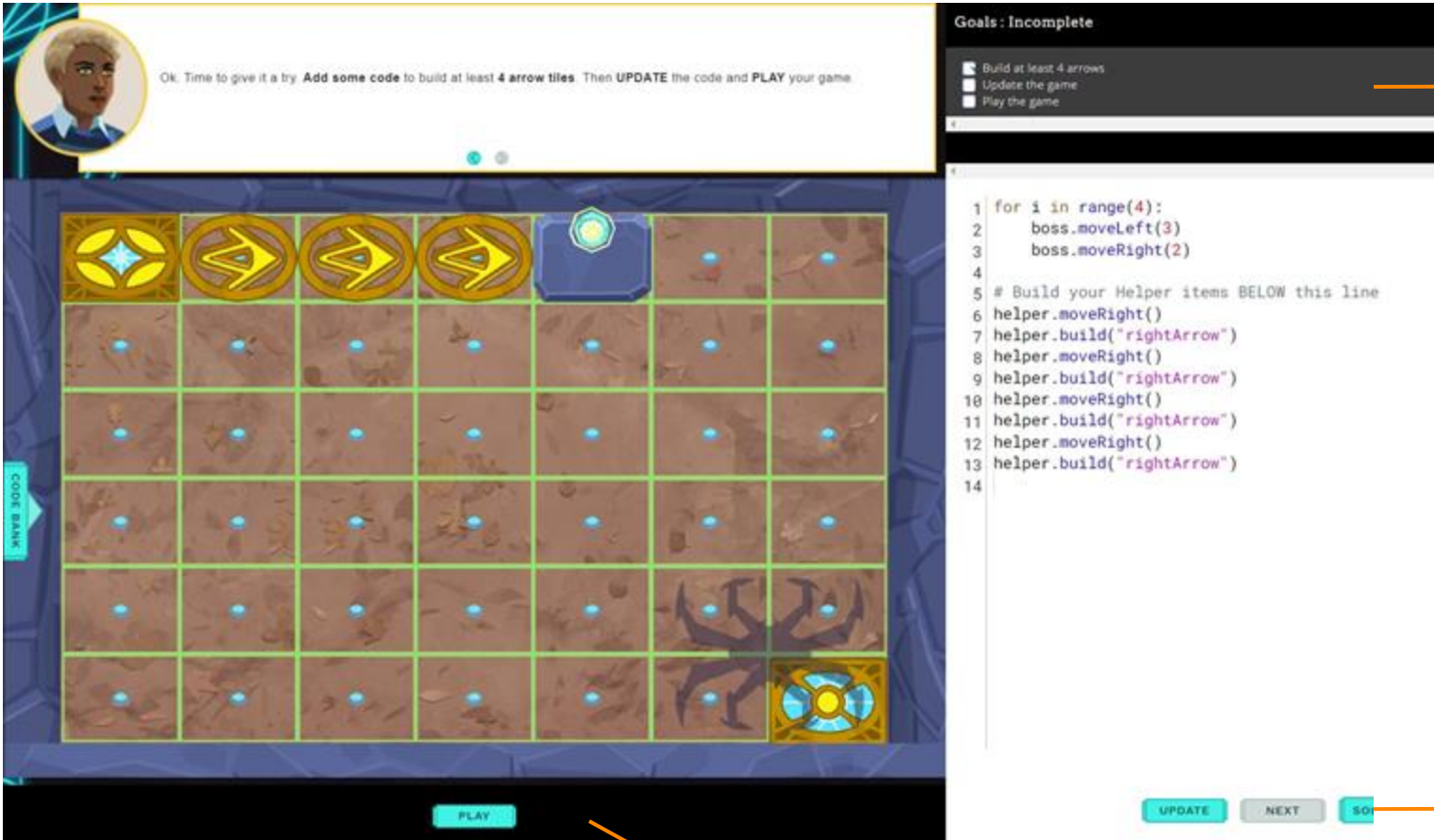
Capstone Example

- [Capstone project example](#)
- [Advanced capstone example](#)

Rubric & Expectations

- [Capstone Rubric - Student](#)
- [Capstone Rubric - Teacher](#)

Capstone Project Interface

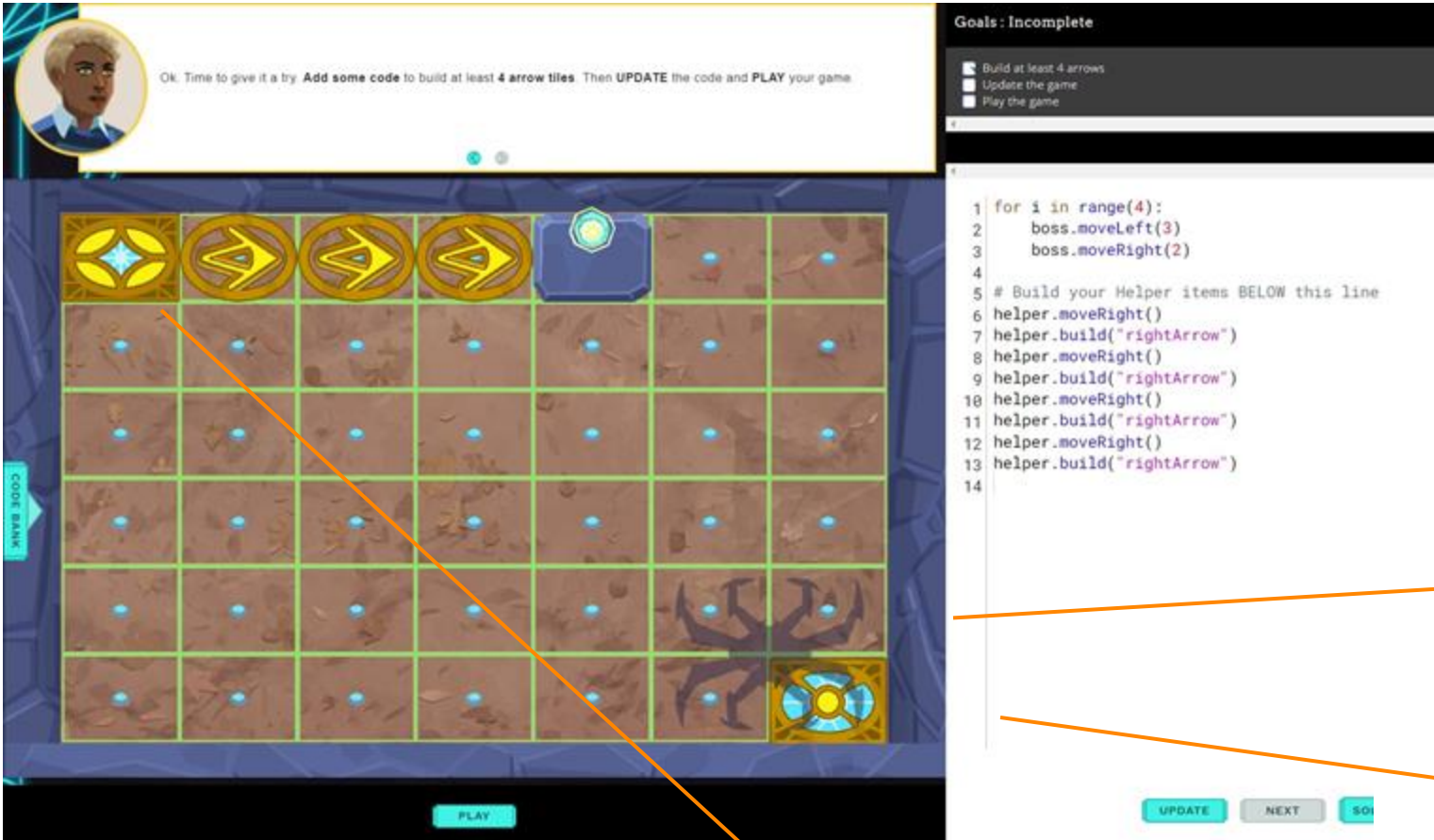


These **goals** will change as you complete each stage of the capstone project.

The **Update Button** will update the game screen to show everything the helper has built in your code.

The **Play Button** lets you play your game.

Capstone Project Interface



If the player runs into the **boss**, then it returns to the start tile.

To win the game, the player needs to get to the **End Tile**.

The **Start Tile** is where the player will always start in your game.

Capstone Project Stages

Here's what you'll be building in each stage of the project



Stage 1: Building Arrows

- Build at least 4 arrows in your game.

Stage 2: Building Obstacles & Items

- Build obstacles and items such as a wall and spikes.

Stage 3: Programming the Boss Loop

- Change the movement of the boss by editing the boss loop in the code.

Stage 4: Adding More Customization

- Customize your game by adding more items, changing the artwork, and using other methods found in the Code Bank.

Check-In





What did you discover?



Chapter 1, Module 1, Lesson 3

What I did:

-How far did you get in Ozaria? What was your favorite part?

What I noticed:

- What did you notice as you played through the game?
- Did you discover or learn something new?

What was challenging?

- Was something confusing about the story or your capstone project?
- Do you need more practice on a specific part of the code or the game?



Lesson

Capstone Project

4



What did we learn last time?



- What are **loops**?
- What are the goals of our capstone project?
- What have you completed so far in your capstone?
- What's going on in the story of **Ozaria**?



Today's Journey



Warm-Up: What makes a good game?



Finish your **Capstone Project**



Get **peer feedback** on your project



Present your project

Game Design



What makes a good game?

Different parts of a game determine whether people will enjoy playing it



Audience

- Consider who you're designing the game for
- Is the audience young kids, adults, or your peers?
- Is your game for casual players or intense players?

Clear Instructions

- Provide clear rules & a clear goal

Challenging

- What makes your game challenging?
- What makes a game too easy or too hard?
- What makes you want to play again?

Storytelling

- Choosing art and a design that tells a story



Concept Check: Game Design

How do you test and revise your game so that it's more fun to play?



Testing & Feedback

- How do you ask for constructive feedback?
- How do you give constructive feedback?

Revision Options

- Use items to tell a story
- Add items to make the level more challenging

Exemplary Examples

To reach an exemplary score for Design & Creativity and Coding Concepts, make at least 3-4 additions to your capstone project that are similar to the following:



Customize and Add More Elements

- **Change the player** design.
- **Change the avatar** design.
- Build at least **3 walls, 3 spikes, and 3 blocks** for your game.

Add Complexity to the Game

- Program the **boss** to move **over every grid** on the game map.
- Build one of the following designs:
 - Create a **recognizable shape or word** using arrows and obstacles in the game
 - **Create only two paths** that the player can take to get to the goal

Take Time to Reflect

Take some time to reflect on what you've done in this chapter.



What did you do?

- What characters did you interact with in this chapter?
- What were some interesting goals that you achieved?

What did you notice?

- How has your coding changed as you played this chapter?
- What were some new coding concepts that you learned?

What were the challenges?

- What was challenging in this chapter?
- What concepts would you like to practice more?

Entrance Survey (Results)



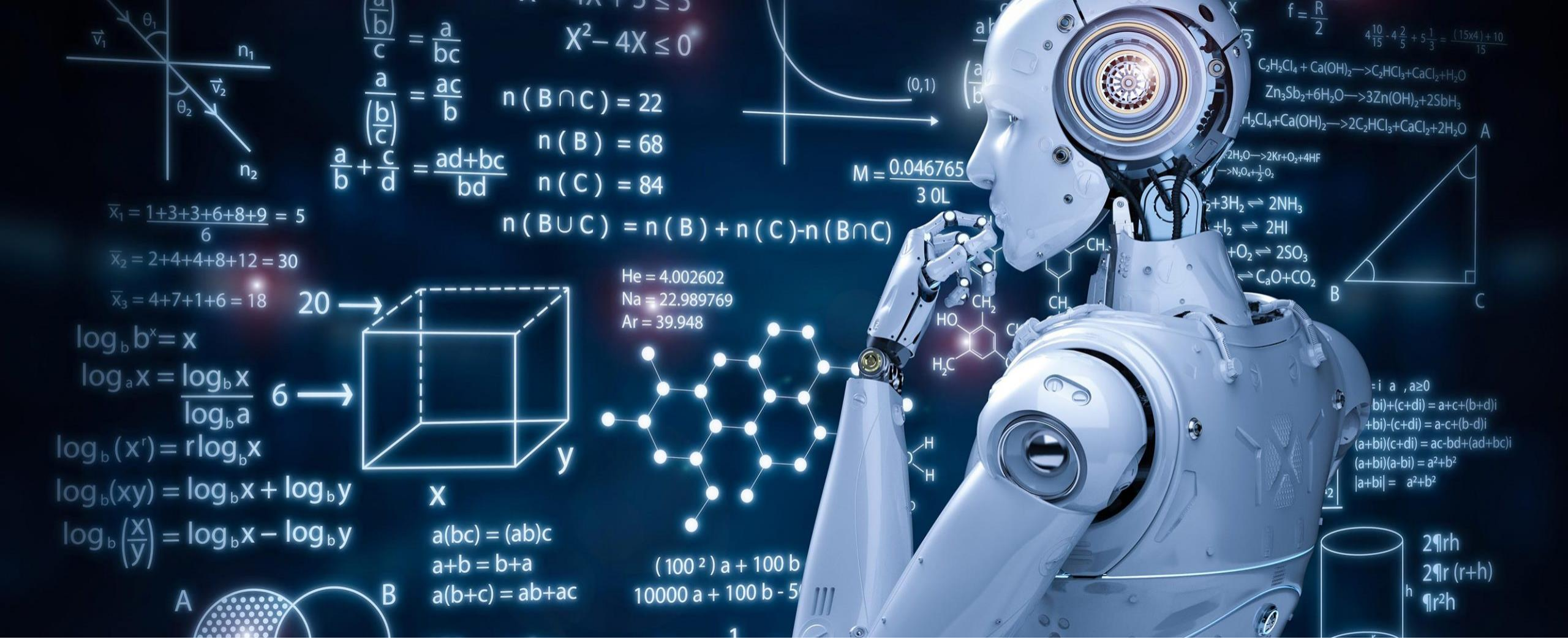
Use of Class Time

- Course is capped at 50 seats
- Each seat costs \$50
- EIP has invested \$2500 to offer this course
- Bring in schoolwork to occupy yourself



Supercomputing Conference

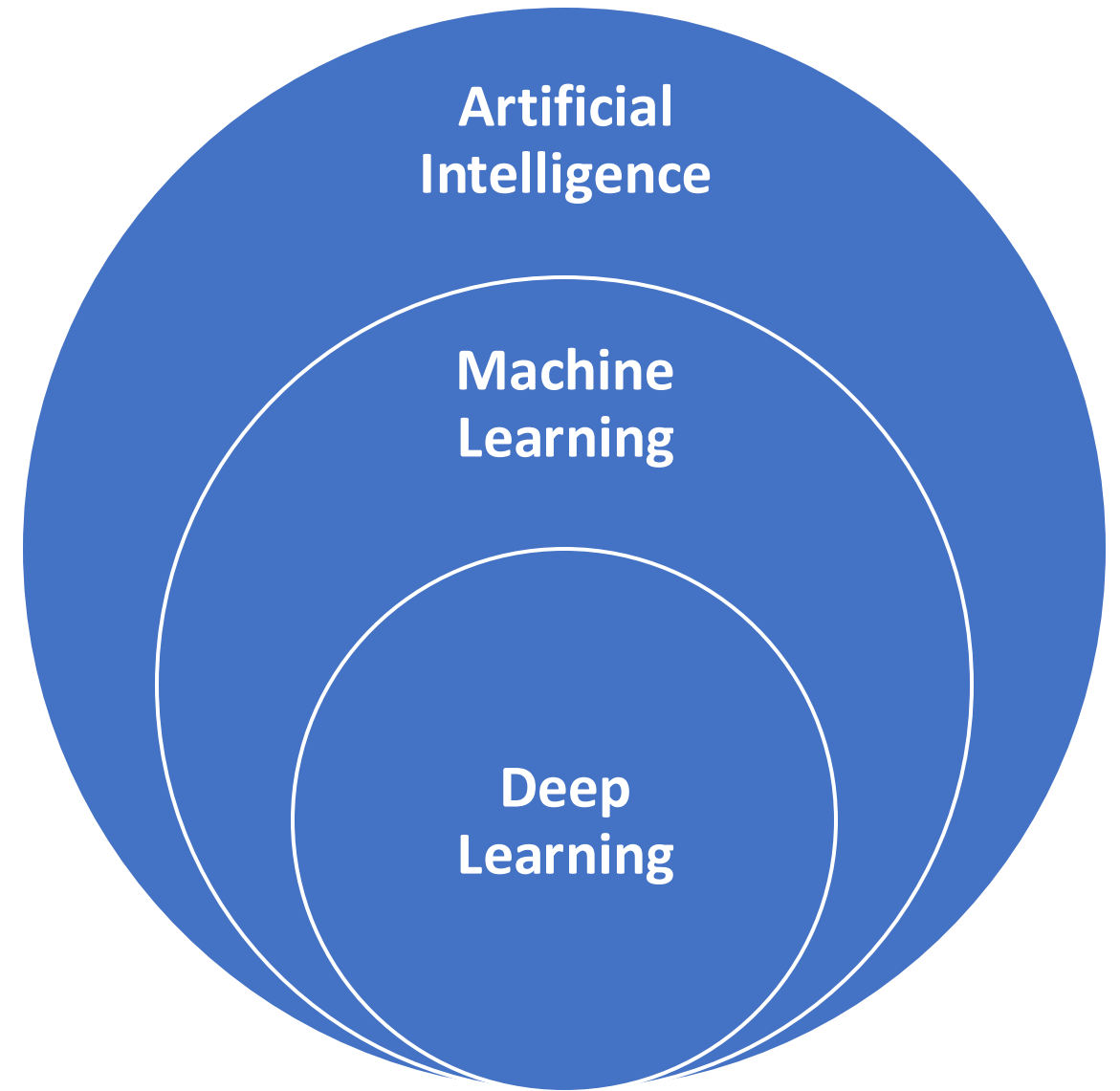




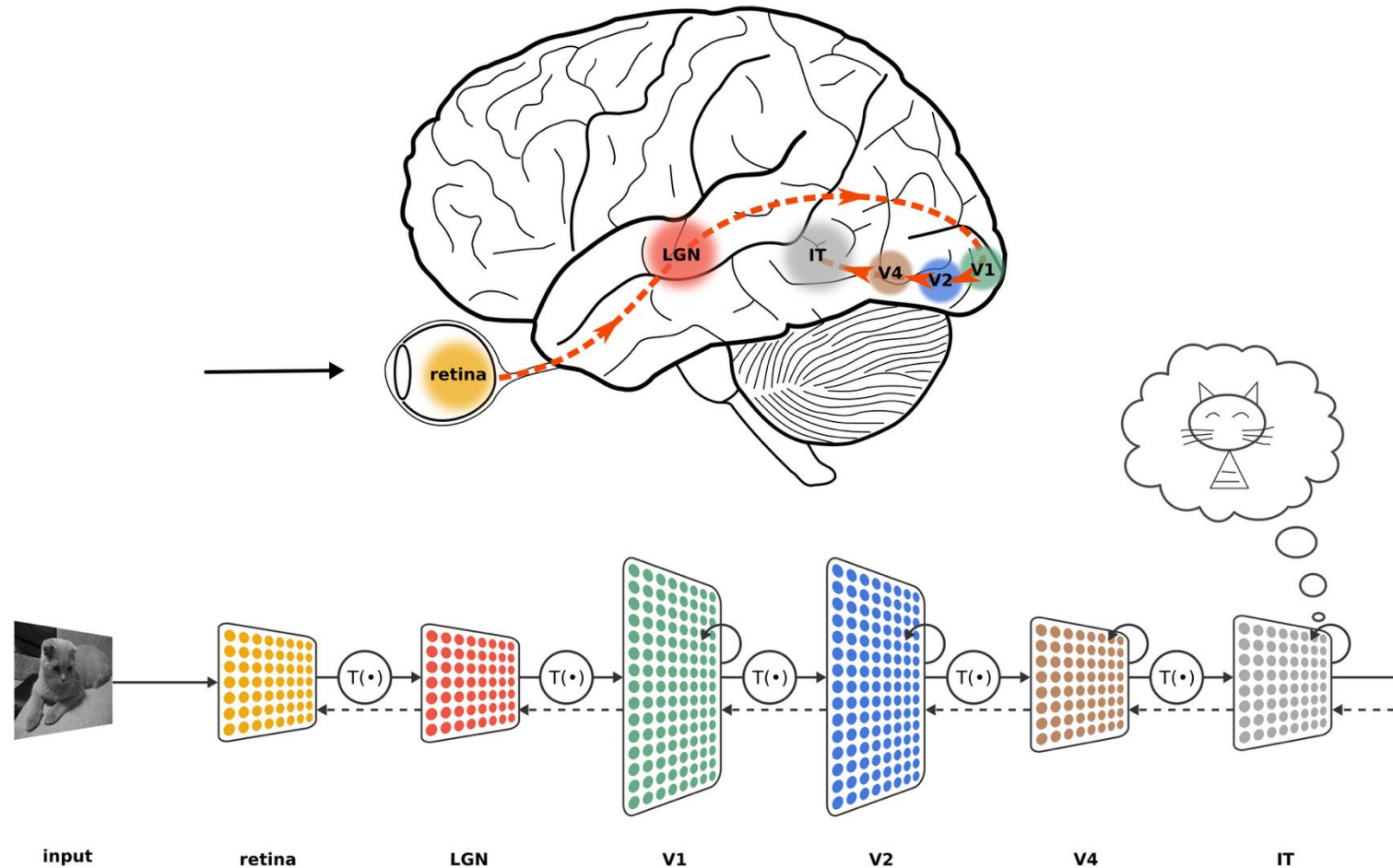
Generative AI

Buzzwords: AI/ML/DL

- AI - Creation of **intelligent machines** that can **perform tasks** that would typically **require human intelligence** to complete.
- ML - teaching computers to **learn** and **make decisions** on their own.
- DL - type of ML that uses multi-layer neural networks to **analyze** and **interpret data** (recognize **patterns**).



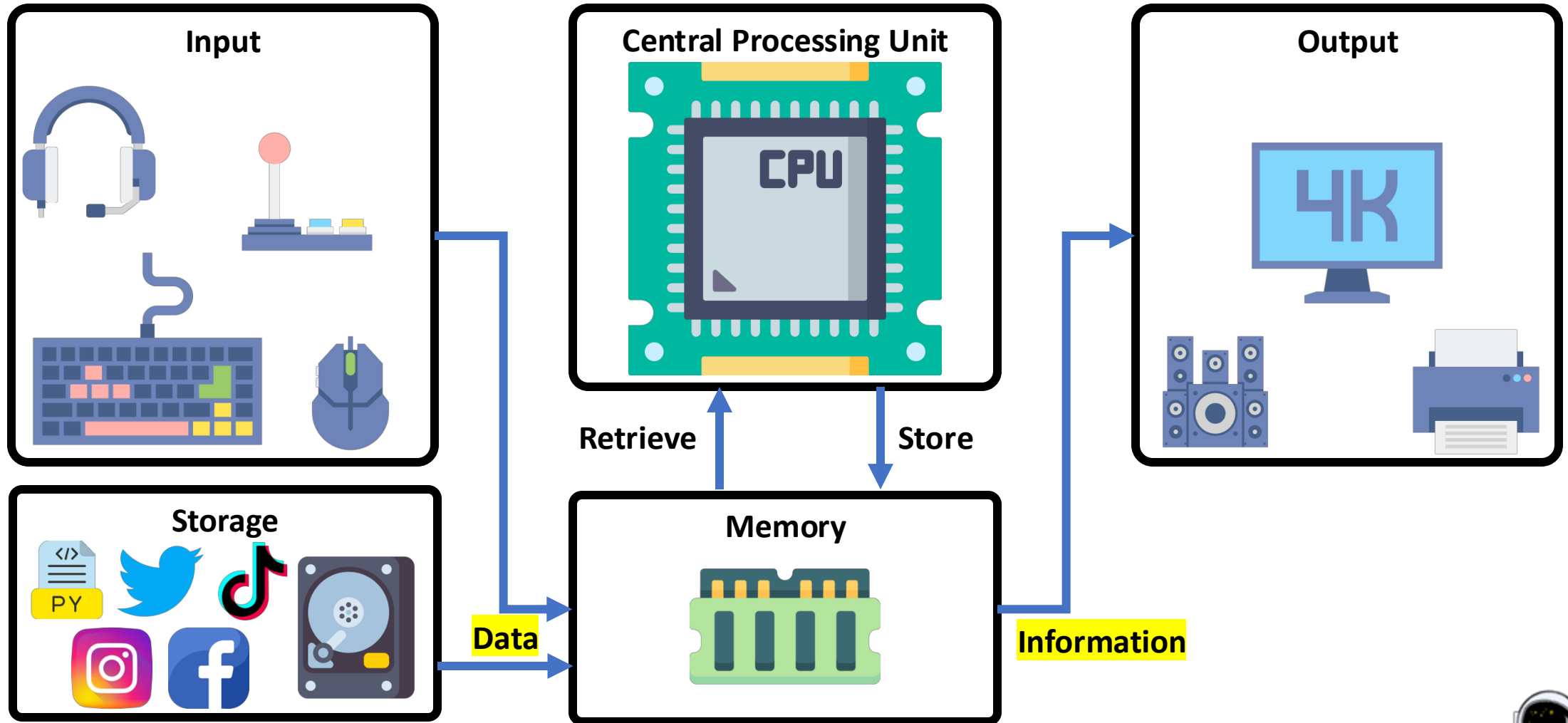
Neural Networks



What is Generative AI?

- Generative AI is a type of artificial intelligence that can **create new content that is similar to the original data it was trained on.**
- It **learns patterns and relationships in data**, and **uses that knowledge to generate new content**, such as images, music, or text.
- It's used in many different applications, including art, music, writing, and video game development.
- It can also be used to create more realistic simulations for scientific research and to **generate synthetic data** for machine learning models.

Data -> Information



Data -> Information -> Knowledge

Data (raw data)	Information (processed and organized)	Knowledge (insight gained)
Census records	Demographics	What will be the average age of the population in 2040?
Weather measurements	Average temperature, rainfall, etc. in a region over a given time range	How will the weather be next week, month, year?
Financial records	Growth rate of a company	Where will this company be in the next five years?
Patient records	Incident rate of a disease	When will COVID-19 end?

Popular Generative AI Tools

- **ChatGPT:** This is a language model that can generate human-like text based on a given prompt.
- **Microsoft 365 Copilot:** AI for Microsoft Office
- **Midjourney/DALL-E:** This is an image generation AI that can generate images from textual descriptions.
- **MuseNet:** This model can generate original music along with multi-instrumental pieces.

ChatGPT

- Interact with this AI model by providing prompts.
- Prompts may describe a task being asked of the model.
- Prompts are inputs to the model.
- Generative content are the outputs from the model.
- The performance (quality of the output) of these models are dependent on the inputs.
- Prompt Engineering - creating specific prompts or inputs that can guide the generative AI model to produce output that is relevant and coherent.

ChatGPT Prompts

- Classify the named entities in this text: George Washington and his troops crossed the Delaware River on December 25, 1776 during the American Revolutionary War.
- Translate this text into Portuguese: Welcome to the Matrix
- Explain this python code:

```
r=int(input("Enter upper limit: "))  
for a in range(2,r+1):  
    k=0  
    for i in range(2,a//2+1):  
        if(a%i==0):  
            k=k+1  
    if(k<=0):  
        print(a)
```


ChatGPT Prompts

- Write a positive Yelp review of Café Renaissance in Vienna, Virginia. Highlighting their grilled halibut dish in almond and lemon butter sauce.
 - More witty and colorful.
 - End with: It was a wonderful way to spend Valentines Day.
 - Also highlight the Escargot à la Bourguignonne appetizer which was cooked in garlic butter parsley and brandy.
- Create a template for a letter of recommendation for Dr. Sarom Leang. Dr. Leang was an Early Identification Program scholar during undergrad. Dr. Leang earned his BS in chemistry and a minor in computer science from George Mason University in 2004 and completed his PhD in quantum chemistry from Iowa State University in 2012.

ig.ft.com/generative-ai



How Do Large Language Models (LLMs) Work?

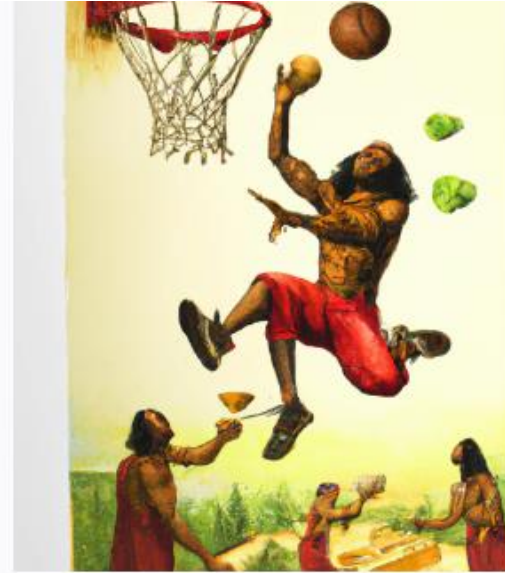
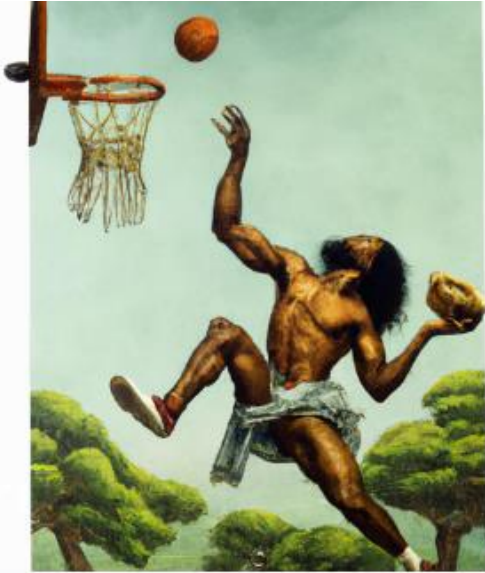
- LLMs are statistical pattern-recognition and prediction systems
- LLMs output the next likely word (“token”) in a sentence (“sequence”)
 - Token: unit of text e.g. word, character. 1 word ~ 0.75 token
 - Sequence: context - section (“window”) of text e.g. sentence, paragraph, book
 - Input into chatGPT is 4096 tokens; Claude 2 is 100K tokens
- The likelihood of the next word appearing is determined by
 - The context in which the words are seen in a larger body of text (“corpus”) and
 - The input to the chat



DALL-E: An abstract image of George Mason on a horse riding into a rainbow sunset



DALL-E: A painting of Michael Jordan slam dunking in the air while eating an avocado



DALL-E: Synthwave George Mason University logo



DALL-E: A photo of an avocado playing basketball with a pomegranate



DALL-E: A painting of dark-skinned female Santa having dinner with dark-skinned female Jesus



Generative AI Inception

- Ask ChatGPT to generate a prompt for DALL-E.

Create an image of a fantastical underwater city, with towering structures and bustling streets, where mermaids and other sea creatures live and work alongside human-like beings. The city should have a distinct aesthetic, with colors and architectural elements that evoke the feeling of being submerged in the depths of the ocean. The image should also include unique transportation methods, such as aquatic vehicles or creatures that are used for travel.



Generative AI Limitations

- Generative AI can produce content that is **incorrect** or **incoherent** (hallucinate).
- Generative models don't understand the content they generate. They **generate responses based on statistical patterns rather than comprehension**, which means they can't answer questions that require understanding or reasoning.
- Generative AI can perpetuate **biases** present in the training data.
- Generative models can be **verbose** and tend to **over generate** content.
- Same prompt may produce **inconsistent** responses using the same model.

Generative AI and Academic Code of Conduct

- Cheating or violating academic code of conduct using generative AI is unethical and has serious consequences.
- Disciplinary action may include failing the assignment or course, or even expulsion from school.
- Long-term consequences may include damaging a student's reputation and future academic and career prospects.
- **Use generative AI responsibly and in accordance with the academic code of conduct.**